5. Running thread · Method [ALG] Procedure: choose a number x ; add 3; multiply by 2; subtract 6. Calculate the result for $x = 5$, then write an expression in x and simplify it. What do you notice?

Ans. For 5: 10. Expression: 2 $(x+3)-6=2x$: we always obtain twice the chosen number.

Link A09: algebraic notation proves what numerical trials suggest.

Sheet 12 Tue. 17 Nov. 2026 Intersection of two lines; Python collinearity function

1. Warm-up · Flash [EQ] Solve 2 $x+1=-x+7$.

Ans. $3x=6$, so $x=2$.

2. Core · Method [GEO] Coordinates of the intersection of $y=2x+1$ and $y=-x+7$.

Ans. $x=2$ then $y=2\times 2+1=5$: $(2;5)$.

Check in the second equation: $-2+7=5$.

3. Core + · Reading code [ALG] What are the outputs of alignes(0,0,1,2,2,4) and alignes(0,0,1,2,2,5) ?

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
```

Ans. True then False : determinants $1\times 4-2\times 2=0$ and $1\times 5-2\times 2=1\neq 0$.

4. Review · Expand [DEV] Expand and simplify 3 $(x-2)$.

Ans. $3x-6$.

5. Running thread · Prove [LIT] Show that the sum of three consecutive integers is a multiple of 3.

Ans. With $n-1$, n, $n+1$: sum $=3n$, multiple of 3.

Choosing the middle term simplifies the calculation.

Real numbers and intervals

Sheet 13 Thu. 19 Nov. 2026 Sets of numbers: $\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$

1. Warm-up · Flash [NUM] Give the decimal expansion of $\frac{3}{8}$ and an approximate value of $5\frac{}{6}$ à 10^{-3} accuracy.

Ans. 0,375 ; 0,833.

2. Core · Method [NUM] Give the smallest set ($\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$) containing each of these numbers: -4 ; 0,25 ; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.

Ans. $-4\in\mathbb{Z}$; $0{,}25\in\mathbb{D}$; $\frac{1}{3}\in\mathbb{Q}$; $\sqrt{2}\in\mathbb{R}$; $\frac{6}{3}=2\in\mathbb{N}$.

The smallest set is the key: $\frac{6}{3}$ is a natural number.

3. Core + · True/false [NUM] True or false? Justify. The number $\frac{1}{3}$ is a decimal number.

Ans. False. If there existed a and n integers with $\frac{1}{3}=\frac{a}{10^n}$, then $10^n=3a$ would be divisible by 3: impossible (the sum of the digits of 10^n is equal to 1).

4. Review · Flash [SUB] Calculate $(x+1)(x-3)$ for $x=2$.

Ans. -3.

5. Running thread · Flash [SOL] How many faces, vertices and edges does a right triangular prism have?

Ans. 5 faces (2 triangles, 3 rectangles), 6 vertices, 9 edges.

Sheet 14 Mon. 23 Nov. 2026 Decimal bounds and rounding

1. Warm-up · Flash [NUM] Round 3.14159 to 10^{-2} accuracy, then to 10^{-3} accuracy.

Ans. 3.14 then 3.142.

The next digit decides: $1<5$ *puis* $5\geqslant 5$.

2. Core · Method [NUM] Bound $\sqrt{2}$ between two decimals differing by 10^{-2}.

Ans. $1{,}41^2=1{,}9881<2<2{,}0164=1{,}42^2$: $1{,}41<\sqrt{2}<1{,}42$.

3. Core + · Reading code [ALG] What is the final value of `k` ? What can be deduced about $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

Ans. $k = 15$: $1{,}4^2 = 1{,}96 < 2$ and $1.5^2 = 2{,}25 \geqslant 2$, therefore $1.4 < \sqrt{2} \leqslant 1{,}5$.

4. Review · Flash [QP] 5 kg of rice cost 12,500 Ar. How much is 8 kg?

Ans. $\frac{12\,500}{5} \times 8 = 20{,}000$ Ar.

5. Running thread · True/false [DEM] True or false? Justify. A quadrilateral whose diagonals have the same length is a rectangle.

Ans. False. Counterexample: an isosceles trapezium has diagonals of equal length without being a rectangle.

Sheet 15 Tue. 24 Nov. 2026 Irrational numbers: $\frac{1}{3}$ is not a decimal, $\sqrt{2}$ is irrational

1. Warm-up · Flash [CAL] Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b an integer as small as possible.

Ans. $\sqrt{12} = 2\sqrt{3}$; $\sqrt{50} = 5\sqrt{2}$.

2. Core · Sorting [NUM] Among $0.333\ldots$; $\sqrt{9}$; $\sqrt{2}$; π ; $\frac{22}{7}$, which ones are rational?

Ans. $0{,}333\ldots = \frac{1}{3}$; $\sqrt{9} = 3$; $\frac{22}{7}$: rational. $\sqrt{2}$ and π are irrational.

$\frac{22}{7}$ is only an approximate value of π, not equal to π.

3. Core + · Reasoning [DEM] Suppose $\sqrt{2} = \frac{p}{q}$ with $\frac{p}{q}$ irreducible, hence $p^2 = 2q^2$. What can be said about the parity of p, then of q ? What contradiction?

Ans. p^2 even, so p even, $p = 2k$; then 4 $k^2 = 2q^2$, $q^2 = 2k^2$: q even. Contradiction with the fraction irreducible.

4. Review · Table [STA] Values 1; 2; 3; 4 with frequencies 5; 8; 4; 3. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Count	5	8	4	3

Ans. Total 20; frequency $\frac{8}{20} = 40\%$; mean 5 $\frac{+16+12+12}{20} = 2{,}25$.

5. Running thread · Flash [REP] Give the coordinates of the reflection of $A(3;-2)$ in the y-axis.

Ans. $(-3;-2)$.

Pitfall: changing the sign of the y-coordinate (reflection in the x-axis).

Sheet 16 Thu. 26 Nov. 2026 Intervals of $\mathbb{R}$: notation and representation

1. Warm-up · Flash [CMP] Compare -3 and -5 then $-\frac{1}{2}$ and $-0{,}6$.

Ans. $-3 > -5$; $-0{,}5 > -0{,}6$.

For negative numbers, the one with the smaller absolute value is the larger.

2. Core · Method [NUM] Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.

Ans. $[-2;5[$; $]3;+\infty[$; $]-\infty;0]$.

The bracket faces inwards if the endpoint is included; ∞ is always excluded.

3. Core + · MCQ [NUM] The intersection $]-\infty;2[\cap [-1;4]$ is:

A. $]2;4]$ **B.** $]-\infty;4]$ **C.** $[-1;2[$ **D.** $[-1;2]$

Ans. C. $[-1;2[$

Intersection: values satisfying both conditions x < 2 *et* $-1 \leqslant x \leqslant 4$.

4. Review · Table [LEC] Travel times (min).